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DETAIL SPECIFICATION

ELECTRODES, WELDING, COVERED: GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers covered welding electrodes.

1.2 Classification. Welding electrodes are furnished in the types, classes, and sizes specified in the specification sheets.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-10430 - Welding Rods and Electrodes; Packaging of

(See ASSIST database for list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AWS A4.3 - Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding

(Copies of this document are available online at https://webstore.ansi.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A3.0M/A3.0 - Standard Welding Terms and Definitions Including Terms for Additive Manufacturing, Adhesive Bonding, Brazing, Soldering, Thermal Cutting, and Thermal Spraying, and Nondestructive Examination

AWS A4.4M - Standard Procedures for Determination of Moisture Content of Welding Fluxes and Welding Electrode Flux Coverings

AWS A5.01M/A5.01 - Welding and Brazing Consumables – Procurement of Filler Metals and Fluxes

AWS A5.02/A5.02M - Specification for Filler Metal Standard Sizes, Packaging, and Physical Attributes

AWS A5.5/A5.5M - Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding

AWS B4.0 - Standard Methods for Mechanical Testing of Welds

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM E10 - Standard Test Method for Brinell Hardness of Metallic Materials

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

ASTM E316 - Standard Test Method for Determination of Iron in Manganese Ores by Hydrogen Sulfide Reduction-Dichromate Titrimetry

ASTM E350 - Standard Test Methods for Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron, and Wrought Iron

(Copies of these documents are available online at www.astm.org.)

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC 60974 - Arc Welding Equipment

(Copies of this document are available online at www.iec.ch.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA EW 1 - Electric Arc-Welding Power Sources

(Copies of this document are available online at www.nema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Electrodes furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Materials.

3.3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Asbestos. The electrodes and their containers shall be asbestos-free (see 3.12.1 and 4.3.4.7).

3.3.3 Material from previous production runs. Materials from previous production runs shall be excluded from any new production run to be offered for acceptance, unless the materials have been determined to be acceptable for meeting all requirements of this specification (see 4.6.1).

3.4 Coverings of electrodes.

3.4.1 Concentricity. The coverings on all sizes of electrodes shall be concentric to the extent that maximum core-plus-one covering dimension shall not exceed the minimum core-plus-one covering dimension by more than that specified in AWS A5.02/A5.02M (see 4.6.2).

3.4.2 Uniformity. Core wire and coverings shall be free of defects that would interfere with uniform performance of the electrodes (see 4.6.3).

3.4.3 Dielectric strength. The coverings of electrodes at room temperature and in the dry condition, that is, as removed from freshly opened containers, shall have a dielectric strength that shall insulate against a difference potential of 110 volts (V), 60 hertz (Hz), alternating current (AC) (see 4.6.4).

3.4.4 Flaking and cracking of covering. The sum of the surface area of the core wire exposed by flaking and cracking of the covering shall not exceed that permitted for arc ends (see 3.4.6.1, 4.6.5, and 4.6.7).

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3.4.5 Slag characteristics. The slags deposited by the coverings shall be removable with hand tools (i.e., slag removal shall not require power tools although power tools may be used) from the weld deposits. Slag characteristics shall be such that grinding during test plate preparation shall not be required for slag removal (see 4.6.6).

3.4.6 Exposed core.

3.4.6.1 Arc ends. The arc end of each electrode shall be in accordance with AWS A5.02/A5.02M (see 4.6.7).

3.4.6.2 Grip ends. The grip end of each electrode shall be in accordance with AWS A5.02/A5.02M (see 4.6.7).

3.4.7 Electrode identification. Each electrode shall be identified (imprinted) in accordance with AWS A5.02/A5.02M except that the MIL grade shall be indicated on each electrode. When specified (see 6.2), each electrode shall also be marked (in addition to the requirements of AWS A5.5/A5.5M) with the heat, lot, or other controlled marking code (see 4.6.8).

3.4.8 Covering composition. The chemical composition of the coverings except iron powder and moisture is at the manufacturer's discretion. When moisture and iron content testing is required by the specification sheet, the required limits shall be as specified in the specification sheet (see 3.1, 4.6.9.1, and 4.6.9.2).

3.5 Core wires of electrodes.

3.5.1 Core wire dimensions. Standard core wire diameters and lengths of electrodes shall be as specified in the specification sheet. The diameter and length of the core wire shall not vary from that specified by more than the limits in accordance with AWS A5.02/A5.02M (see 4.6.10).

3.5.2 Alloy identity. The material shall be of the type specified when tested in accordance with 4.6.11.

3.6 Weld quality. Electrodes shall deposit groove welds in accordance with class 1 criteria of MIL-STD-2035 when using the nondestructive test methods specified in the specification sheet (see 3.1, 4.6.12.1, 4.6.12.2, and 4.6.12.3).

3.7 Mechanical properties. Mechanical properties shall be as specified in the specification sheet (see 4.6.13).

3.8 Chemical composition of deposited weld metal. Chemical composition shall be as specified in the specification sheet (see 4.6.14 and 6.3.4).

3.9 Diffusible hydrogen. The diffusible hydrogen content of the deposited weld metal shall be as specified in the specification sheet (see 3.1 and 4.6.15).

3.10 Hardness. Hardness of the weld metal shall be as specified in the specification sheet (see 3.1 and 4.6.16).

3.11 Container requirements. Unit packs (containers) of electrodes shall be in accordance with MIL-W-10430 preservation level A, C, or commercial (see 6.2), or AWS A5.02/A5.02M, as specified (see 6.2). Additional requirements shall be as specified (see 4.6.17 and 6.2).

3.12 Container marking requirements. Markings for electrodes, rods, and flux shall be in accordance with MIL-W-10430. Additional requirements shall be as specified (see 4.6.17 and 6.2).

3.12.1 Asbestos-free. Each unit container shall be marked "Asbestos-Free" (see 4.6.17).

3.12.2 Lot identification. Lot identification shall be accomplished by marking each unit and shipping container with a unique manufacturer's control number that is traceable to the original manufacturer's records. If these control numbers are encoded, the interpretive key shall be furnished to the consignee. The control number shall be in a position that can be easily located by the purchaser (see 4.6.17). This control number shall be in a position that permits ready location by the consignee.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (see table I):

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

TABLE I. Qualification and conformance inspections.

Requirement description	Requirement paragraph	Verification paragraph	Verification type [1/]	Qualification	Conformance [2/]
Asbestos	3.3.2	4.3.4.7	Certificate (when specified) Verification statement.	---	X
Material from previous production runs	3.3.3	4.6, 4.6.1	VE	---	X
Covering of electrodes	3.4	---	---	X	X
Concentricity	3.4.1	4.6, 4.6.2	DE [3/]	X	X
Uniformity	3.4.2	4.6, 4.6.3	VE or other [3/]	X	X
Dielectric strength	3.4.3	4.6, 4.6.4	Test	X	---
Flaking and cracking of covering	3.4.4	4.6, 4.6.5	Test	X	---
Slag characteristics	3.4.5	4.6, 4.6.6	Test	X	X
Exposed core	3.4.6	---	---	X	X
Arc ends	3.4.6.1	4.6, 4.6.7	DE [3/]	X	X
Grip ends	3.4.6.2	4.6, 4.6.7	DE [3/]	X	X
Electrode identification	3.4.7	4.6, 4.6.8	VE [3/]	---	X
Covering composition (moisture and iron)	3.4.8 [5/]	4.6, 4.6.9	Test	When specified in the specification sheet.	
Core wires of electrodes	3.5	---	---	X	X
Core wire dimensions (diameter and length)	3.5.1	4.6, 4.6.10	DE [3/]	X	X
Alloy identity	3.5.2	4.6, 4.6.11	Test	X	X
Weld quality	3.6	4.6, 4.6.12	---	RT, MT, and VT required when specified in the specification sheet. [4/]	

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TABLE I. Qualification and conformance inspections – Continued.

Requirement description	Requirement paragraph	Verification paragraph	Verification type [1/]	Qualification	Conformance [2/]
Mechanical properties	3.7 [5/]	4.6, 4.6.13	Test	X	X
Chemical composition of deposited weld metal [6/]	3.8 [5/]	4.6, 4.6.14	Test	X	X
Diffusible hydrogen	3.9 [5/]	4.6, 4.6.15	Test	When specified in the specification sheet.	
Hardness	3.10 [5/]	4.6, 4.6.16	Test	When specified in the specification sheet.	---
Container	3.11	4.6, 4.6.17	VE or other [3/]	---	X
Container marking	3.12	4.6, 4.6.17	VE or other [3/]	---	X
Asbestos-free	3.12.1	4.6, 4.6.17	VE or other [3/]	---	X
Lot identification	3.12.2	4.6, 4.6.17	VE or other [3/]	---	X

NOTES:

[1/] The designations DE and VE are dimensional examination and visual examination, respectively.

[2/] A Certificate of Conformance (CoC) is required per 4.3.4.

[3/] These items fall within the examination sampling requirements specified as "visual and dimensional examinations for electrode characteristics" herein (see 4.3.3.2 and 4.4).

[4/] The designations RT, MT, and VT are radiographic testing, magnetic particle testing, and visual testing, respectively.

[5/] Limits shall be as specified in the specification sheet.

[6/] Changes in chemical formulations or aim values shall be reported to NAVSEA in advance of production implementation, who will determine if requalification is required in cases where the changes are found to adversely affect design criteria or performance (see 6.3.4).

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the conditions specified herein. Environmental conditions shall be ambient laboratory conditions, unless otherwise permitted by 4.6 or the referenced documents therein.

4.2 Qualification inspection. Qualification inspection shall include the examinations and the tests of 4.4 (see 6.3).

4.2.1 Change approval. Once a formulation is developed and qualified, and the aim values (composition validated by qualification testing) for each major alloying element have been established, changes to the formulation, aim values, construction, formulation, materials, design, or manufacturing process of qualified products may require requalification (see 6.3). This information is to be reported to NAVSEA, who will determine if requalification is required in cases where the changes are found to adversely affect design criteria or performance (see 6.3).

4.3 Conformance inspection. Conformance inspection shall include the examinations and the tests of 4.4.

4.3.1 General. Lot classification shall be as defined in 4.3.2.1 and 4.3.2.2. Each lot of electrodes shall be inspected as specified herein and in the specification sheet. For each lot of material, sufficient samples shall be selected to perform the required tests.

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4.3.2 Lot classifications.

4.3.2.1 AWS A5.01M/A5.01, classification C4. When AWS classification C4 or A is specified in the specification sheet, the lot classification shall be in accordance with AWS A5.01M/A5.01, classification C4.

4.3.2.2 AWS A5.01M/A5.01, classification C3. When AWS C3 is specified in the specification sheet, the lot classification shall be in accordance with AWS A5.01M/A5.01, classification C3.

4.3.3 Sampling. Each lot of material offered for acceptance shall be sampled for conformance inspection.

4.3.3.1 Sampling for container, container marking, and lot identification examination. Unless otherwise specified (see 6.2), sampling of each lot for verification of conformance to the container, container marking, and lot identification requirements of section 3 shall be in accordance with MIL-W-10430 (sampling for preservation, packing, and marking).

4.3.3.2 Sampling for visual and dimensional examination for electrode characteristics. Sample electrodes shall be selected either from filled unit packages or from the production line after the baking operation except where otherwise specified in the specification sheet. The unit packages required for this examination may be selected from the sample of packages used for 4.3.3.1. If sample electrodes are selected from filled unit packages or cans, for single sampling, the total sample shall be in accordance with Schedule A of table II (for electrode sizes ⅛-inch and larger) and Schedule B (for electrode sizes 3⁄32-inch and smaller). Approximately the same number of electrodes shall be selected from each of the unit packages or groups of cans (six cans per group). Lot size shall be expressed in pounds. Sample size shall be the number of electrodes to be examined. A minimum number of unit packages shall be selected to permit sampling of ten electrodes maximum from each 50-pound package or from each 10-pound package. If the sample electrodes are selected from the production line, the total sample shall be equivalent to the above and the electrodes shall be selected through the production period so that all parts of the lot are represented. The electrodes selected shall be identified as to type, wet batch, lot, size, and other available information such as contract or order number being filled. The maximum allowable number of defects shall be as specified (see 6.2 and 6.9). If in any sample the number of electrodes that do not conform to this specification exceeds the maximum number of allowable defects specified (see 6.2 and 6.9), this shall be cause for rejection of the lot. Instructions for defective materials shall be as specified (see 6.2 and 6.10).

4.3.3.2.1 Other means of examination. Where 4.6 (also see table I) specifies other means as an alternative to visual examination, the sampling requirements of 4.3.3.2 shall apply.

TABLE II. Sampling for visual and dimensional examinations.

Lot size (pounds of electrodes)	Sample size (number of electrodes to be examined)	
	Schedule A	Schedule B
2 to 8	2	2
9 to 15	2	2
16 to 25	3	3
26 to 50	5	5
51 to 90	5	5
91 to 150	8	8
151 to 280	13	13
281 to 500	13	13
501 to 1,200	20	20

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TABLE II. Sampling for visual and dimensional examinations – Continued.

Lot size (pounds of electrodes)	Sample size (number of electrodes to be examined)	
	Schedule A	Schedule B
1,201 to 3,200	32	32
3,201 to 10,000	32	32
10,001 to 35,000	50	50

4.3.3.3 Sampling for other examinations and tests. An additional quantity of samples shall be selected as specified in 4.3.3.2 to complete the required testing and the remaining examinations other than the visual and dimensional examinations for electrode characteristics.

4.3.4 Certification of conformance. Unless otherwise specified (see 6.2), a certificate of conformance (CoC) shall be prepared for each lot of material offered for acceptance in accordance with the lot definition as specified in the specification sheet. The CoC shall be in accordance with the requirements herein and the specification sheet. A CoC or equivalent test certificate (see 4.3.4.6) shall be furnished to the consignee with the shipment and shall meet the following requirements.

4.3.4.1 Test results. The CoC shall include quantitative results for specified tests herein and the specification sheet, as applicable, for chemical, mechanical, covering moisture, diffusible hydrogen, total iron content, and hardness tests. The CoC shall also include qualitative results of nondestructive tests, wet-mix equivalency test (if applicable), and other inspections and tests. The cause for any retest shall be reported for information and the test results that failed to meet those specified requirements shall be reported.

4.3.4.1.1 Diffusible hydrogen. The test results shall be reported in accordance with AWS A4.3.

4.3.4.2 Stress relief. A record of any stress-relief treatment (if applicable) shall be reported.

4.3.4.3 Welding information. Welding process and operational parameters for welded test assemblies shall be reported.

4.3.4.4 Other information. The manufacturer, distributor, customer, order number, specification, MIL-type, core heat number, lot number, and size (nominal diameter and length) shall be identified.

4.3.4.5 Certification statement. The CoC shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce an electrode that meets the requirements. The CoC shall be signed by a responsible representative of the manufacturer.

4.3.4.6 Test certificates. Where test certificates issued by the manufacturer contain the above information (see 4.3.4.1 through 4.3.4.5), a separate CoC will not be required.

4.3.4.7 Asbestos-free. A CoC shall be submitted when specified (see 3.3.2 and 6.2).

4.4 Examinations and tests. Samples selected in accordance with 4.3.3, as applicable, shall be subject to the examinations and tests for qualification and conformance inspections as designated in table I and as specified in the specification sheet.

4.4.1 Procedures and methods. Procedures and methods shall be as specified in 4.6.

4.4.2 Unsatisfactory test results. Retest, when performed, shall be in accordance with 4.8.

4.5 Test weld fabrication. When welds are required for qualification and conformance inspection tests of deposited weld metal, the fabrication of the weld shall be as specified below and in the specification sheet.

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4.5.1 Welding machines. For both qualification and conformance testing of electrodes, welding machines used to supply power shall be of the following types: (a) variable voltage direct current (DC) motor-generator type, (b) variable voltage or constant voltage DC rectifier welder, (c) variable voltage or constant voltage DC inverter, or (d) an AC welding transformer or an AC inverter. Welding power supplies shall be in accordance with NEMA EW 1 or IEC 60974. Multiple process machines are acceptable provided that they are properly set up. The power source shall be appropriate for the process to be tested and shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 V.

4.5.2 Welding parameters. Welding parameters shall be as specified in the specification sheet.

4.5.3 Base plate material. The base plate material used in the preparation of welded joints shall be as specified in the specification sheet.

4.5.4 Instrumentation. Voltage shall be measured between the electrode holder and the test plate. Welding current and arc voltage measurements shall be made using analog or digital meters. RMS meters shall be used where any AC welding is performed. Tong meters shall not be used unless they are RMS-type and used for AC welding. Instrument calibration records shall be traceable to calibration performed in accordance with ISO/IEC 17025 or equivalent standard.

4.5.5 In-process excavation of groove welds. Grinding, burring, machining, or other excavation during welding of a test plate shall be limited to weld starts and stops, correction of operator errors including use of incorrect welding parameters, and, as necessary, contouring in way of the bead to be subsequently deposited to avoid a welding defect or allow deposition without adjusting the target welding parameters. Excavation to avoid a defect or allow deposition at the target heat input shall only be performed against a previously deposited bead and not against the base plate. The amount, location, and method of excavation shall be recorded by the welder on the test plate work sheet record. Under no circumstances shall grinding or other excavation be performed for the purpose of altering the bead position in an attempt to alter weld metal mechanical properties.

4.6 Procedures and methods. Unless otherwise specified in the specification sheet, procedures and methods for tests and examinations for conformance to the section 3 requirements shall be in accordance with this paragraph. Test weld fabrication shall be in accordance with 4.5. Retest, when performed, shall be in accordance with 4.8.

4.6.1 Pre-production inspection of equipment. Prior to the start of a new production run, where a different electrode type is to be produced, inspection shall be performed to ensure that equipment and process lines are purged of material from the previous production run that do not meet the requirements herein, the requirements in the specification sheet, or conditions for qualification for the new production run. This inspection shall be performed for all equipment used in manufacturing operations where the material is not segregated and positively identified; for example, cutting to length, extruding, and baking (see 3.3.3).

4.6.2 Concentricity. Examination for concentricity shall be in accordance with AWS A5.02/A5.02M (see 3.4.1).

4.6.3 Uniformity. Verification of uniformity shall be by visual examination or by any other means demonstrated to ensure compliance with the requirement specified in 3.4.2.

4.6.4 Dielectric strength. Dielectric strength of coverings shall be determined by the method shown on figure 1. Methods other than the one shown, yielding the required results, may be used as alternates if such methods are acceptable to the Government.

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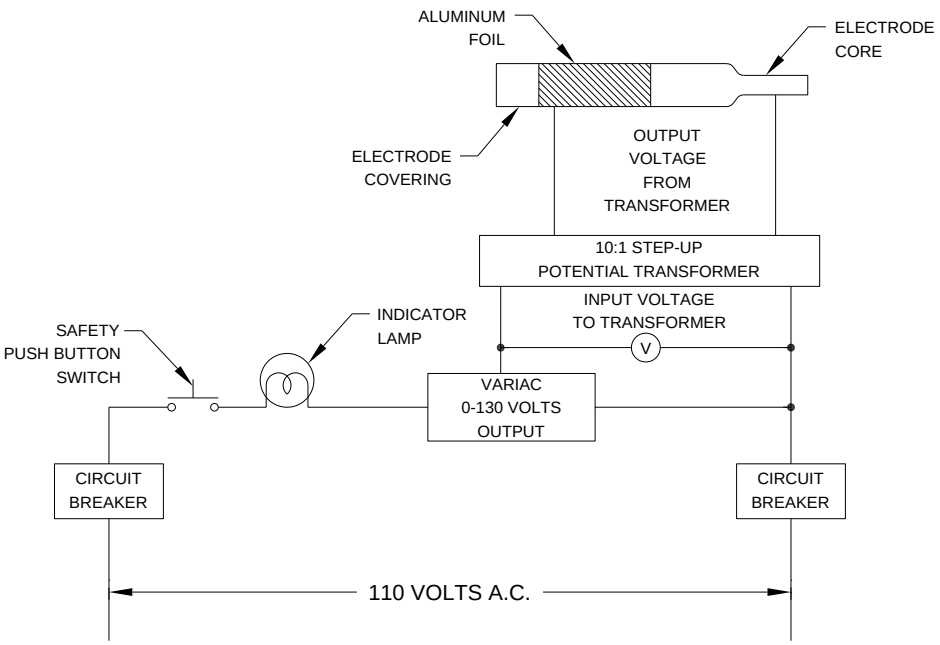


FIGURE 1. Circuit for dielectric strength determination.

NOTES:

1. Wrap conducting aluminum foil around 6-inch length of electrode.
2. Place electrode with grip-end on one lug and foil covered section on other.
3. Close circuit breaker; press safety button and slowly rotate variac control.
4. Record maximum voltage before breakdown.
5. Dielectric strength is recorded voltage, multiplied by 10 (transformer ratio).
6. When dielectric strength is below 300 V, connect appropriate voltmeter across electrode terminals.

4.6.5 Flaking and cracking of covering. Six electrodes shall be tested for flaking and cracking tendency upon restarting and during welding in accordance with 4.6.5.1 and 4.6.5.2, respectively. The polarity shall be the same as required by the specification sheet for the MIL-type electrode being tested. The welding current used shall be in the same range as required by the applicable specification sheet for groove welds. The welding shall consist of depositing a bead weld on the plate surface.

4.6.5.1 Flaking and cracking upon restart. Three electrodes shall be tested. The covering shall be grooved to the core wire completely around the electrode at the midpoint of the length. The groove shall be prepared by grinding with the edge of a grindstone (intersection of flat side of grindstone with its periphery). The core wire shall be uncovered at the root of the groove to form a land 1/16 inch minimum along the length of the electrode. The core wire shall not be undercut. The electrode shall be consumed to the root of the groove in the covering when welding shall be stopped without fusing coating beyond the groove. The half-length electrode shall be removed immediately from the electrode holder and placed upon a flat steel plate to cool until it can be held comfortably in the bare hand. After cooling, the electrode shall be inserted in the electrode holder and welding resumed. After the restart, the half-length electrode shall be consumed for 1 inch of its length. Welding shall then be stopped, and the arc end of the electrode shall be examined for conformance to the requirements of 3.4.4.

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4.6.5.2 Flaking and cracking during welding. Three electrodes shall be tested. The electrode shall be consumed to a stub not over 2 inches in length. If flaking or cracking is observed during welding, the welding shall be stopped immediately and the arc end of the electrode shall be examined for conformance to the requirements of 3.4.4. If no flaking or cracking is observed during welding, the 2-inch stub length of electrode, which is normally discarded, shall not be examined.

4.6.6 Slag characteristics. Verification of slag characteristics shall be determined upon visual examination during the preparation of the groove weld as specified in the specification sheet (see 3.4.5).

4.6.7 Electrode grip and arc ends examination. Grip end and arc end dimensions shall be verified to ensure compliance with the requirements specified in 3.4.6.

4.6.8 Electrode identification examination. Verification of electrode identification shall be by visual examination (see 3.4.7).

4.6.9 Covering composition.

4.6.9.1 Covering moisture. The covering moisture content shall be determined using methods and procedures in accordance with AWS A4.4M (see 3.4.8).

4.6.9.2 Total iron content. The total iron content of the covering, as specified in the applicable specification sheet, shall be made by wet chemical or spectrographic method. In the case of dispute, the referee method shall be in accordance with ASTM E316 (see 3.4.8).

4.6.10 Diameter and length of core wire examination. Diameter and length dimensions of core wire shall be verified to ensure compliance with the requirements specified in 3.5.1.

4.6.11 Alloy identity. Alloy identity methods and procedures shall be in accordance with AWS A5.01M/A5.01 Department of Defense (DoD) supplementary requirements (see 3.5.2 and 6.3.5).

4.6.12 Nondestructive testing of groove welds.

4.6.12.1 Radiographic testing, RT (soundness). Groove welds shall be RT inspected in accordance with T9074-AS-GIB-010/271, using sensitivity level 2-2T (see 3.6).

4.6.12.2 Visual testing (VT). Visual testing of welds shall be performed in accordance with T9074-AS-GIB-010/271 (see 3.6).

4.6.12.3 Magnetic particle testing (MT). MT inspection of welds shall be performed in accordance with T9074-AS-GIB-010/271 (see 3.6).

4.6.13 Mechanical tests. When required in the specification sheet, tensile, Charpy V-notch (CVN), and bend test specimens shall be prepared and tested in accordance with AWS B4.0 (see 3.7).

4.6.13.1 Additional CVN specimen requirements. The CVN specimens shall be in the T-L orientation and centered in the weld joint to ensure that the notch is completely contained in weld metal.

4.6.13.2 Rounding-off procedures for tensile testing. Rounding-off procedures shall be as specified in 4.7.

4.6.13.3 Check mechanical property tolerances. Check mechanical property tolerances shall be as specified in 4.9.

4.6.14 Chemical analysis. Chemical analyses of weld metal shall conform to the following procedures (see 3.8 and 6.3.4):

a. Samples for chemical analysis of the weld metal shall be obtained for each size of electrode using the base plate material specified and the types of current specified for that classification and size.

b. Samples for chemical analysis may be obtained from the weld pad specified in (c) below, or any other weld deposit provided it produces results equivalent to those obtained from the weld pad. In case of dispute, samples shall be taken from the weld pad.

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c. The weld pad shall be deposited in layers in the flat position. The width of each pass in each layer shall be 1½ to 2½ times the diameter of the core wire. After depositing each layer, the pad shall be immersed in water (temperature unimportant) for 15 to 45 seconds and then dried before welding is resumed. The surface of each layer shall be free of all foreign matter. The completed weld pad and the location from which the sample for analysis is taken shall conform to the requirements in table III.

d. Chemical analysis shall be made by any accepted analytical method agreed upon by the supplier and purchaser. In case of dispute, the procedure in ASTM E350 shall be the referee method, unless otherwise specified in the detail specification.

TABLE III. Weld pad dimensions.

Electrode size (inch)	Minimum pad size (inch)	Minimum distance of sample from surface of base plate (inch)
3/32	1 by 1 by ½	¼
⅛	1½ by 1½ by ⅝	5/16
5/32		
3/16		
7/32	2 by 2 by ¾	⅜
¼		
5/16		

4.6.15 Diffusible hydrogen. The diffusible hydrogen level of the deposited weld metal shall be determined by the analytical method of gas chromatography or by the method of collection over mercury in accordance with AWS A4.3 (see 3.9).

4.6.16 Hardness. When hardness testing is required in the specification sheet, the hardness of the weld metal shall be measured by the Brinell method in accordance with ASTM E10. The hardness shall be measured at approximately the center of the cross-sectional area of groove welds. Rockwell hardness numbers converted to Brinell hardness numbers shall not be used for decisive tests (see 3.10).

4.6.17 Container, container marking, and lot identification examination. Container, container marking, and lot identification examinations shall be accomplished by visual examination, or by any other means demonstrated to ensure compliance with requirements specified in 3.11 and 3.12.

4.7 Rounding-off procedure. For purposes of determining conformance with this specification, an observed or calculated value shall be rounded off to the nearest 1,000 pounds per square inch for tensile and yield strength, and to the “nearest unit” in the last right-hand place of figures used in expressing the limiting value for other values in accordance with the rounding-off method specified in ASTM E29 (see 4.6.13.2).

4.8 Unsatisfactory test results. Retest shall be in accordance with the following (see 4.6).

4.8.1 Retests. When any original required conformance test or examination representing a lot of material fails to meet specification requirements, the lot may be retested at the manufacturer’s discretion. Retest is not applicable to pre-production inspection of equipment (see 4.6.1), slag characteristics (see 4.6.6), or container, container marking, and lot identification examinations (see 4.6.17). If the manufacturer chooses to retest the lot, the retests shall be as follows:

a. If the test failed due to one (or more) of the conditions listed in 4.8.1.2, then the lot shall be retested according to the requirements in 4.8.1.3.

b. If the test failed due to conditions other than those listed in 4.8.1.2, then the lot shall be retested according to the requirements in 4.8.1.4.

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4.8.1.1 Test set. All the specimens in a single test, whether the test involves a single specimen (such as for RT) or multiple specimens (such as for CVN impact tests), shall be considered a "set".

4.8.1.2 Testing and specimen preparation error retests. Retests to correct errors in testing and test specimen preparation are permitted under the following conditions:

a. Improperly fabricated test welds that have not been fabricated in accordance with the applicable specification or the weld procedure for the product under test.

b. Improperly prepared specimens that are not in compliance with applicable standards.

c. Test equipment malfunctions in the process of conducting a test.

d. Flaws in test specimens that are not indicative of inferior or defective product. Flaws at weld starts and stops at the ends of the test assembly, lack of fusion at the root, and slag due to operator technique are normal weld artifacts and do not count as flaws indicative of inferior or defective products. Cracks, clustered porosity, flakes, and ruptures are flaws that may be indicative of inferior or defective products and shall not qualify a lot for a testing or specimen preparation error retest.

4.8.1.3 Testing or specimen preparation error retest requirements. Retests for conditions listed in 4.8.1.2 shall be as follows:

a. For each original test that failed due to one (or more) of the conditions in 4.8.1.2, only one testing or preparation error retest is permitted.

b. A testing or preparation error retest shall consist of one replacement specimen for each specimen that failed due to the conditions in 4.8.1.2. Note for VT, MT, and RT, the specimen is the entire weld. If the testing error is due to a malfunction of MT or RT equipment, or to a flaw not indicative of inferior or defective products (see 4.8.1.2.d) removed by grinding or other mechanical methods within 1/16 inch of the surface, the original test assembly may be retested.

c. Results of all retests and reasons for all retests shall be reported on the CoC.

d. Retests of weld metal (e.g., mechanical properties) shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

e. Retests of the product (e.g., diffusible hydrogen, moisture, and chemistry pads) shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

f. If the results of the testing or preparation error retest do not satisfy the applicable requirements, then a conformance retest shall be performed per 4.8.1.4.

4.8.1.4 Conformance retest requirements. Retests for failure to meet test requirements shall be as follows:

a. For each original test that failed due to conditions other than those listed in 4.8.1.2, only one conformance retest is permitted.

b. A conformance retest shall consist of repeating the test that failed two times (i.e., two-for-one). Note that a mechanical property test such as tensile or impact toughness involves a set of test specimens and, therefore, two full sets of specimens for each test that failed are required under this two-for-one rule. Results of all retests and reasons for all retests shall be reported on the CoC.

c. Retests of weld metal shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

d. Retests of product shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

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e. Unless the welding procedure was determined to be the root cause of the original test failure, any replacement test weld should employ the same welding procedure as the original test weld. If a new procedure is used to correct the root cause of the original test failure, the retest shall include all tests conducted for the original weld. Two-for-one is only required for the test that failed. All future lots of the material shall use the new procedure. The reason for changing procedure shall be justified to the customer and included on the CoC.

f. If the results of any conformance retest do not satisfy the applicable requirements, then the lot shall be rejected.

4.9 Check mechanical testing tolerances. Unless otherwise specified (see 6.2), repeating the mechanical property conformance tests as part of receipt inspection by the purchaser shall employ the following check mechanical property testing tolerances to determine acceptability of the product to the requirements in the specification sheet.

a. Yield strength test results (rounded off in accordance with 4.7) may be 3 percent higher than the specified maximum or 3 percent lower than the specified minimum requirements of the specification sheet.

b. Ultimate tensile strength test results (rounded off in accordance with 4.7), when required, may be 3 percent lower than the specified minimum requirements of the specification sheet.

c. Elongation test results shall not be lower than 93 percent of the minimum requirement specified in the specification sheet.

d. Charpy V-notch test results (both the average and the individual minimum value) may be lower than the requirements specified in the specification sheet by 10 percent.

4.10 Inspection of packaging and marking. Packaging and marking shall be inspected for compliance with the contract (see 5.1, 6.2, and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2 and 6.5). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The products covered by this specification include shielded metal arc welding (SMAW) electrodes intended for welding steels in the as-welded and stress-relieved condition (MIL-DTL-22200/1, MIL-DTL-22200/8, and MIL-DTL-22200/12), austenitic chromium-nickel alloy (MIL-DTL-22200/2), nickel, nickel-alloy, chromium-copper alloy (MIL-DTL-22200/3), and copper-nickel alloy (MIL-DTL-22200/4). This specification contains requirements above and beyond those of commercial specifications for use in critical military applications. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet.

c. Whether heat, lot, or other marking codes are required for individual electrodes (see 3.4.7).

d. Level of preservation required in accordance with MIL-W-10430 or if AWS A5.02/A5.02M is required (see 3.11).

e. Any additional container requirements in addition to MIL-W-10430 or AWS A5.02/A5.02M requirements (see 3.11).

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f. Any additional container label requirements (see 3.12).

g. Inspection conditions, if other than specified (see 4.1.1).

h. Whether a sampling plan other than that specified in MIL-W-10430 for container, container label, or lot identification examination is required (see 4.3.3.1).

i. Maximum number of allowable defects required for visual and dimensional inspections (see 4.3.3.2).

j. Instructions for defective materials (see 4.3.3.2 and 6.10).

k. If a CoC is not required (see 4.3.4).

l. Whether a CoC indicating electrodes and containers are asbestos-free is required (see 4.3.4.7).

m. When check mechanical property tolerances are not permitted for retest as part of receipt inspection (see 4.9).

n. Packaging requirements (see 4.10, 5.1, and 6.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 22200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Materials. The coating formulation, method of processing, composition of the deposited weld metal, and procedure for moisture control between baking and packaging are to form a part of the qualification submittal. Any subsequent change thereto may require requalification or additional tests and are subject to the approval of the NAVSEA.

6.3.2 Special instructions. Qualification is authorized only to manufacturers of welding electrodes. When applying for test authorization, or after tests have been authorized and when samples are submitted, the manufacturer furnishes the information specified in 6.3.2.1 and 6.3.2.2. (NOTE: This information, together with test results obtained with the electrode sample, forms a part of the qualification test; all information will be held in confidence by the Government.)

6.3.2.1 Classification C4. For covered electrode lots in accordance with AWS A5.01M/A5.01, classification C4 (see 4.3.2.1), the following is furnished for approval by NAVSEA:

a. The lengths of the electrodes and the diameter of coating and core wire.

b. Type and class under which approval is desired.

c. Composition of core wire and covering(s) in terms of nominal percentages for each constituent. (Information must be maintained at the manufacturer’s plant for Government audit purposes.)

d. Composition of the deposited weld metal.

e. Recommended amperages for each weld test. (Information must be maintained at the manufacturer’s plant for Government audit purposes.)

f. Brand name.

6.3.2.2 Classification C3. For covered electrode lots in accordance with AWS A5.01M/A5.01, classification C3 (see 4.3.2.2), in addition to 6.3.2.1, the following is furnished for approval by NAVSEA:

a. Chemical composition control limits in core wire of each MIL-type electrode. (Information must be maintained at the manufacturer's plant for Government audit purposes.)

b. Method of determining core wire chemistry.

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c. Production line methods used to produce electrodes from chemically-controlled core wire. (Information must be maintained at the manufacturer's plant for Government audit purposes.)

d. Procedure for control of moisture content between baking and packaging.

e. Percent of allowable variation (disclosed) from the standard (not disclosed) for each chemical element in the covering mixture of each MIL-type electrode. (Information must be maintained at the manufacturer's plant for Government audit purposes.)

f. Method of determining covering mixture chemistry.

g. Production line methods used to produce electrodes from chemically controlled covering mixture. (Information must be maintained at the manufacturer's plant for Government audit purposes.)

6.3.3 Samples for qualification inspection. Each manufacturer furnishes sample packages and electrodes of the type the manufacturer desires to have inspected. Quantities are requested by the qualifying activity or laboratory and are selected at one time in the presence of the Government representative and so indicated. The manufacturer indicates on the sample whether the electrodes and packages were produced on a laboratory or experimental scale, or on a production scale.

6.3.4 Chemical analysis. The chemical compositions as specified in the specification sheets are minimum and maximum limits. However, the specification does not imply, and it should not be inferred, that each of the possible chemistry compositions defined by these limits will meet the full requirements of the specification, for example, mechanical properties. At the time of qualification, the aim values and typical ranges must be reported for each element that represent expected variations due to manufacturing tolerances.

6.3.5 Alloy identity methods and procedures. Alloy identity methods and procedures must be submitted to NAVSEA as part of qualification for approval. NAVSEA must be notified of changes.

6.4 Recommendations for stability of electrodes and packaging. It is recommended that a warranty period be established in the contract as shown in table IV. The stability of the electrodes and of the packaging should be such that electrodes after receipt from the manufacturer will comply with table IV after storage in original unopened containers under roof and on dry platforms. The 6-month or 1-year period should not be taken, by the consignee, as a limit on the useful life of the electrodes.

TABLE IV. Stability.

Type of unit container	Time period
Class 5 of MIL-W-10430	Up to 6 months
Classes 1 and 2 of MIL-W-10430	Up to 1 year

6.5 Packaging recommendations. The following minimum packaging recommendations are provided. Alternate or additional requirements may be necessary as determined by the purchaser.

6.5.1 Protection against damage. Electrodes are to be suitably packaged to ensure against damage during shipment and storage under normal conditions.

6.5.2 Packaging and marking. Electrodes should be preserved to level A, C, or commercial and packed at level A, B, C, or commercial, as specified in the contract, in accordance with MIL-W-10430. Electrodes, rods, and flux should be marked in accordance with MIL-W-10430. Additional requirements should be as specified (see 6.2).

6.5.3 Packaging materials. Unless otherwise specified (see 6.2), packaging materials should be in accordance with MIL-W-10430.

6.5.3.1 Loose-fill materials. The use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation or storage. If cushioning is required, specify the use of cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material.

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6.5.3.2 Navy fire-retardant materials. When specified, Navy fire-retardant requirements for lumber, plywood, and fiberboard used in packaging should be in accordance with MIL-W-10430.

6.6 Inspection after delivery. Post-delivery inspection of electrodes, rods, flux, and packaging to determine conformance to this specification and for acceptance thereof is determined by the Command or Agency concerned. Post-delivery inspection testing of electrodes intended for critical applications, such as submarine structures, should be in accordance with the applicable fabrication document.

6.7 Mixed-up material recommendations. Subsequent to shipment, if a heat or lot is found to contain material of a type or class different from that specified, the contractor should notify all other customers, who procured to the requirements herein and in the applicable specification sheet and received material from that heat or lot, of the condition. The contractor should be held responsible for re-inspection to ensure that any unused material from the same heat or lot is correct, and should replace all incorrect material at no additional charge.

6.8 Definitions. For the purpose of this specification, the welding terms and definitions contained in AWS A3.0M/A3.0 must apply.

6.8.1 Command or agency concerned. Command or agency Government, prime contractor, or authorized representative who has design or acquisition responsibility acting under contract to the Government.

6.8.2 Contractor. The contractor is the seller under the contract or purchase order which incorporates this specification.

6.8.3 Manufacturer. The manufacturer is the actual processor of the welding electrodes and rods engaged in the final cleaning, spooling, cutting to length, affixing rod identification, marking, and packaging operations.

6.9 Maximum allowable number of defects for visual and dimensional examination of electrode characteristic samples. These recommendations do not apply to sampling for other examinations and tests (see 4.3.3.3). Table V provides the maximum number of defects for samples selected in accordance with 4.3.3.2 and table II herein. These maximum numbers are the same as was provided in the previous revision of this specification.

TABLE V. Sampling for visual and dimensional examination for electrode characteristics.

Lot size, pounds of electrodes	Sample size, number of electrodes for examination		Maximum number of defective electrodes for acceptance of the lot	
	Schedule A	Schedule B	Schedule A	Schedule B
2 to 8	2	2	0	0
9 to 15	2	2	0	0
16 to 25	3	3	0	0
26 to 50	5	5	0	0
51 to 90	5	5	0	0
91 to 150	8	8	0	1
151 to 280	13	13	1	2
281 to 500	13	13	1	2
501 to 1,200	20	20	2	3
1,201 to 3,200	32	32	3	5
3,201 to 10,000	32	32	3	5
10,001 to 35,000	50	50	5	7

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6.10 Defective material replacement recommendations. When a lot is found to be acceptable when applying any fixed AQLs, lot tolerance percent defectives, or other types of fixed levels of defects, the defective electrodes in the lot found to be defective should be replaced with non-defective electrodes prior to shipment. In no case should known defective material be shipped.

6.11 Certificate of conformance (CoC) considerations. The CoC as required in this specification is an essential element for assuring that the material being purchased is in conformance with the specification. The CoC content, as required herein, provides valuable information regarding the welding parameters used for the required tests, expected performance, actual weld metal chemistry, whether the product was subjected to a stress-relief heat treatment, information regarding test failures, and other information that links the CoC to the manufacturer, purchaser, specification, lot, etc. Navy technical and safety program documentation typically require certain information included in the prescribed CoC for use as objective quality evidence (OQE) demonstrating specification conformance and compliance as well as traceability for materials used within prescribed ship systems. Such documentation includes, but is not limited to, SUBSAFE, Level I, Deep Submergence Scope of Certification (DSSOC) and ship hull, machinery, and piping systems fabrication documents. A CoC may also be required by customs. The CoC also provides beneficial information in support of hull structure and component failure analyses, tracking problematic heats or lots of material throughout the supply system or onboard ships, and material, component, and specification improvements. These added benefits are possible because of the unique tracking information provided within the CoC and the unique set of mechanical properties and chemical content provided for each lot of material.

6.12 Subject term (key word) listing.

Core wire

Shielded metal arc welding

Welding electrode

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 84

Review activities:
Army – MR
Navy – YD

Preparing activity:
Navy – SH
(Project 3439-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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